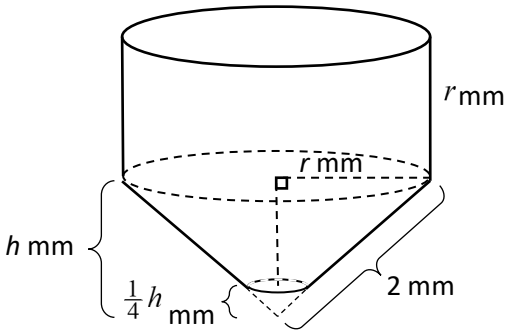
RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 1 (Zeta)
Topic(s): Calculus I
(Differentiation, Applications, Maclaurin and Graphing (Solutions)

1 HCI Prelim 9758/2023/01/Q11

[The volume of a right circular cone of radius r and height h is $\frac{1}{3}\pi r^2 h$.]



A student makes a printer nozzle for his self-built 3D printer. The printer nozzle consists of a hollow inverted right circular cone of negligible thickness with radius r mm where $0 < r < 2$, height h mm and slanted edge 2 mm joined to a hollow cylinder of negligible thickness with radius r mm and height r mm. A height of $\frac{1}{4}h$ mm is cut off from the vertex of the cone for the nozzle opening (see diagram).

The volume of the printer nozzle is V mm^3.

(a) Show that $V = \pi r^3 + \frac{21}{64}\pi r^2\sqrt{4-r^2}$. [3]

The student wants V to be a maximum.

(b) It is given that $r = r_1$ gives the maximum value of V. Show that r_1 satisfies the equation $4537r^4 - 18736r^2 + 3136 = 0$. [4]

(c) Show that one of the positive roots of the equation in part **(b)** does not give a stationary value of V. Suggest a reason why this value is a solution to the equation in part **(b)** even though it does not give a stationary value of V. [3]

(d) Given that r_1 is the largest positive root of the equation in part **(b)**, state the value of r_1 and find the corresponding value of h. (You need not show that your answer gives a maximum.) [1]

(e) With reference to the value of r_1 found in part **(d)**, comment on the practicality of having a maximum volume for the printer nozzle. [1]

Solution:

(a) By Pythagoras' Theorem,

$r^2 + h^2 = 2^2$

$\therefore \quad h = \sqrt{4 - r^2}$ (reject $h = -\sqrt{4 - r^2}$ since $h > 0$)

Let r_0 be radius of cut off cone.

Using similar $\triangle$s, $\frac{r_0}{r} = \frac{1}{4} \Rightarrow r_0 = \frac{r}{4}$

$$\therefore V = \pi r^2(r) + \frac{1}{3}\pi r^2 h - \frac{1}{3}\pi r_0{}^2(\frac{1}{4}h)$$
$$= \pi r^3 + \frac{1}{3}\pi r^2\sqrt{4 - r^2} - \frac{1}{192}\pi r^2\sqrt{4 - r^2}$$
$$= \pi r^3 + \frac{21}{64}\pi r^2\sqrt{4 - r^2} \quad \text{(shown)}$$

(b)

$$\frac{\mathrm{d}V}{\mathrm{d}r} = 3\pi r^2 + \frac{21}{64}\pi\left[2r\sqrt{4 - r^2} + r^2(\frac{1}{2\sqrt{4 - r^2}})(-2r)\right]$$
$$= 3\pi r^2 + \frac{21}{64}\pi r\left[2\sqrt{4 - r^2} - \frac{r^2}{\sqrt{4 - r^2}}\right]$$
$$= 3\pi r^2 + \frac{21}{64}\pi r\left(\frac{8 - 3r^2}{\sqrt{4 - r^2}}\right)$$

At maximum value of V, $\frac{\mathrm{d}V}{\mathrm{d}r} = 0$,

$$\therefore \ 3\pi r^2 + \frac{21}{64}\pi r\frac{8 - 3r^2}{\sqrt{4 - r^2}} = 0$$

Take out common factors

$$3\pi r\left[r + \frac{7}{64}\left(\frac{8 - 3r^2}{\sqrt{4 - r^2}}\right)\right] = 0$$
$$r + \frac{7}{64}\left(\frac{8 - 3r^2}{\sqrt{4 - r^2}}\right) = 0 \quad \text{(since } 0 < r < 2\text{, and so } r \neq 0\text{)}$$
$$r = \frac{7}{64}\left(\frac{3r^2 - 8}{\sqrt{4 - r^2}}\right)$$

Square both sides

$$64r\sqrt{4 - r^2} = 21r^2 - 56$$
$$4096r^2(4 - r^2) = (21r^2 - 56)^2$$
$$16384r^2 - 4096r^4 = 441r^4 - 2352r^2 + 3136$$
$$\therefore \ 4537r^4 - 18736r^2 + 3136 = 0 \quad \text{(shown)}$$

(c) Using GC, since $r > 0$,

$r = 0.41806129085388$ or $r = 1.9886743863834$

When $r = 0.41806129085388$, $\dfrac{\mathrm{d}V}{\mathrm{d}r} = 3.294435715 \neq 0$

Hence $r = 0.41806129085388$ does not give a stationary value of V.

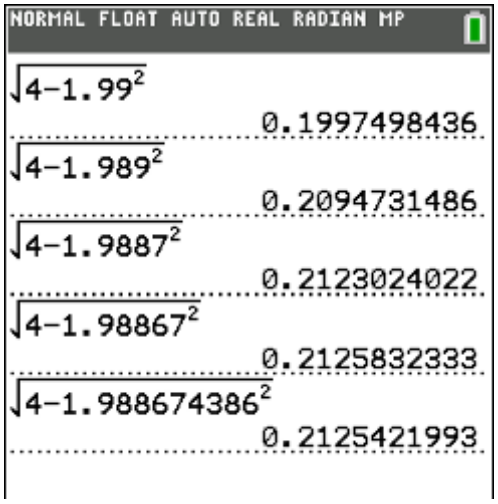
The squaring of the equation to remove the square root created additional roots to the equation. These additional roots may not give rise to stationary values of V.

Note: For example, let's say $p = -2$. If we square both sides this equation, we obtain $p^2 = 4$. And solve this quadratic equation gives $p = -2$ or 2. The solution $p = 2$ is extraneous and was not in the original equation. So, squaring both sides "destroys" the sign information. One should always verify the answers in the original equation whenever one square both sides.

(d) $r_1 = 1.99$ (3 s.f.)

$h = \sqrt{4 - r_1^2} = 0.21254 = 0.213$ (3 s.f.)

Note:



Notice that if we use $r_1 = 1.99$ (3 s.f.), we will get h = 0.200 (3 s.f.) instead of 0.213 (3 s.f.). So, it is important to use a more accurate intermediate values before getting your final answer.

(e) The value of $r_1 = 1.99$ is almost the same length as the slant height 2 of the inverted cone. This means that the inverted cone is essentially flat and non-existent, leaving the printer nozzle in the shape of a cylinder only. Hence it is not realistic to have a maximum volume for the printer nozzle.

Ideal measurements/values/conditions based on theoretical calculations may not be practical or feasible in real life.

2 SAJC Prelim 9758/2025/P2/Q5

A curve C has parametric equations

$$x = \tan\theta - \cot\theta, \quad y = \tan\theta + \cot\theta, \quad \text{for } 0 < \theta < \frac{\pi}{4}.$$

(i) Show that $\dfrac{\mathrm{d}y}{\mathrm{d}x} = -\cos 2\theta$. [3]

The line L is the tangent to C at point P with parameter p.

(ii) Show that the equation of L is $y + (\cos 2p)x = 2\sin 2p$. [3]

Let Q and R be the points where L cuts the x-axis and y-axis respectively.

(iii) Given that p increases at a constant rate of 0.2 radians per second, find the exact rate of change of the area of ΔOQR when $p = \dfrac{\pi}{6}$, where O is the origin. [4]

Solution:

(i)

$x = \tan\theta - \cot\theta \qquad y = \tan\theta + \cot\theta$

$\dfrac{\mathrm{d}x}{\mathrm{d}\theta} = \sec^2\theta + \operatorname{cosec}^2\theta \qquad \dfrac{\mathrm{d}y}{\mathrm{d}\theta} = \sec^2\theta - \operatorname{cosec}^2\theta$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}\theta} \div \frac{\mathrm{d}x}{\mathrm{d}\theta}$$
$$= \frac{\sec^2\theta - \operatorname{cosec}^2\theta}{\sec^2\theta + \operatorname{cosec}^2\theta}$$
$$= \frac{\dfrac{1}{\cos^2\theta} - \dfrac{1}{\sin^2\theta}}{\dfrac{1}{\cos^2\theta} + \dfrac{1}{\sin^2\theta}}$$
$$= \frac{\sin^2\theta - \cos^2\theta}{\sin^2\theta + \cos^2\theta}$$
$$= \sin^2\theta - \cos^2\theta$$
$$= -\cos 2\theta \text{ (Shown)}$$

Alternatively,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\sec^2\theta - \operatorname{cosec}^2\theta}{\sec^2\theta + \operatorname{cosec}^2\theta} = \frac{\sec^2\theta\left(1 - \dfrac{\operatorname{cosec}^2\theta}{\sec^2\theta}\right)}{\sec^2\theta\left(1 - \dfrac{\operatorname{cosec}^2\theta}{\sec^2\theta}\right)}$$
$$= \frac{1 - \cot^2\theta}{1 + \cot^2\theta}$$
$$= \frac{1 - \cot^2\theta}{\operatorname{cosec}^2\theta}$$
$$= \frac{1}{\operatorname{cosec}^2\theta} - \cos^2\theta$$
$$= \sin^2\theta - \cos^2\theta$$
$$= -\left(\cos^2\theta - \sin^2\theta\right)$$
$$= -\cos 2\theta \text{ (Shown)}$$

(ii)

Coordinates of $P = (\tan p - \cot p, \tan p + \cot p)$ and

Gradient of line $L = \cos 2p$

Equation of line L:

$$y-(\tan p+\cot p)=-\cos 2p(x-(\tan p-\cot p))$$
$$y+(\cos 2p)x=\tan p+\cot p+\cos 2p\tan p-\cos 2p\cot p$$
$$y+(\cos 2p)x=\tan p(1+\cos 2p)+\cot p(1-\cos 2p)$$
$$y+(\cos 2p)x=\tan p(2\cos^2 p)+\cot p(2\sin^2 p)$$
$$y+(\cos 2p)x=\frac{\sin p}{\cos p}(2\cos^2 p)+\frac{\cos p}{\sin p}(2\sin^2 p)$$
$$y+(\cos 2p)x=2\sin p\cos p+2\sin p\cos p$$
$$y+(\cos 2p)x=2\sin 2p \quad \text{(shown)}$$

(iii)

Given $\frac{\mathrm{d}p}{\mathrm{d}t}=0.2$.

x-intercept of L (Point Q):

When $y=0$, $(\cos 2p)x=2\sin 2p$

$$x=2\tan 2p$$

y-intercept of L (Point R):

When $x=0$, $y=2\sin 2p$

Note: Since $0<\theta<\frac{\pi}{4}$, so $0<p<\frac{\pi}{4}$

$$\Rightarrow\ 0<2p<\frac{\pi}{2}$$

So, $\tan 2p>0$ and $\sin 2p>0$

Area of OQR, $A=\frac{1}{2}(2\sin 2p)(2\tan 2p)=2\tan 2p\sin 2p$

$$\frac{\mathrm{d}A}{\mathrm{d}p}=2\left(\tan 2p(2\cos 2p)+(\sin 2p)2\sec^2 2p\right)$$
$$=4\left(\sin 2p+\tan 2p\sec 2p\right)$$

At $p=\frac{\pi}{6}$,

$$\frac{\mathrm{d}A}{\mathrm{d}p}=4\left(\sin\left(\frac{\pi}{3}\right)+\tan\left(\frac{\pi}{3}\right)\sec\left(\frac{\pi}{3}\right)\right)=4\left(\frac{\sqrt{3}}{2}+2\sqrt{3}\right)=4\left(\frac{5}{2}\right)\sqrt{3}=10\sqrt{3}$$

$$\frac{\mathrm{d}A}{\mathrm{d}t}=\left(\frac{\mathrm{d}A}{\mathrm{d}p}\right)\left(\frac{\mathrm{d}p}{\mathrm{d}t}\right)=10\sqrt{3}\times 0.2=2\sqrt{3}\ \text{units}^2/\text{s}$$

3 ACJC Prelim 9758/2021/01/Q1 (modified)

(a) A function is defined as $\mathrm{f}(x)=\dfrac{1-\mathrm{e}^{x}}{\mathrm{e}^{x-2}}$.

Describe a sequence of transformations that transforms the graph of $y=\mathrm{e}^{x}$ onto the graph of $y=\mathrm{f}(x)$. [5]

(b) A function is defined as $\mathrm{g}(x)=\cos 2x$.

(i) Describe a sequence of transformations that transforms the graph of $y=\sin 2x$ onto the graph of $y=\mathrm{g}(x)$. [2]

(ii) Describe a sequence of transformations that transforms the graph of $y=\sin^{2} x$ onto the graph of $y=\mathrm{g}(x)$. [3]

Solution:

(a)

$$\begin{aligned}\mathrm{f}(x)&=\frac{1-\mathrm{e}^{x}}{\mathrm{e}^{x-2}}\\&=\frac{1}{\mathrm{e}^{x-2}}-\frac{\mathrm{e}^{x}}{\mathrm{e}^{x-2}}\\&=\mathrm{e}^{-x+2}-\mathrm{e}^{2}\end{aligned}$$

Let $\mathrm{g}(x)=\mathrm{e}^{x}$. Then, $\mathrm{f}(x)=\mathrm{e}^{-x+2}-\mathrm{e}^{2}=\mathrm{g}\left(2-x\right)-\mathrm{e}^{2}$

1. Translation of 2 units in the negative x-axis direction.
2. Reflection in the y-axis.
3. Translation of e^{2} units in the negative y-axis direction.

1. $y=\mathrm{g}(x)\rightarrow y=\mathrm{g}(x+2)$
2. $y=\mathrm{g}(x+2)\rightarrow y=\mathrm{g}(-x+2)$
3. $y=\mathrm{g}(-x+2)\rightarrow y=\mathrm{g}(-x+2)-\mathrm{e}^{2}$

Alternative 1:

1. Reflection in the y-axis.
2. Translation of 2 units in the positive x-axis direction.
3. Translation of e^{2} units in the negative y-axis direction.

1. $y=\mathrm{g}(x)\rightarrow y=\mathrm{g}(-x)$
2. $y=\mathrm{g}(-x)\rightarrow y=\mathrm{g}(-(x-2))=\mathrm{g}(2-x)$
3. $y=\mathrm{g}(2-x)\rightarrow y=\mathrm{g}(2-x)-\mathrm{e}^{2}$

Alternative 2:

$\mathrm{f}(x)=\mathrm{e}^{-x+2}-\mathrm{e}^{2}=\mathrm{e}^{2}\cdot\mathrm{e}^{-x}-\mathrm{e}^{2}$

Let $\mathrm{g}(x)=\mathrm{e}^{x}$. Then, $\mathrm{f}(x)=\mathrm{e}^{2}\cdot\mathrm{e}^{-x}-\mathrm{e}^{2}=\mathrm{e}^{2}\mathrm{g}\left(-x\right)-\mathrm{e}^{2}$

1. Reflection in the y-axis.
2. Scaling parallel to the y-axis by the scale factor e^{2}.
3. Translation of e^{2} units in the negative y-axis direction.

1. $y=\mathrm{g}(x)\rightarrow y=\mathrm{g}(-x)$
2. $y=\mathrm{g}(-x)\rightarrow y=\mathrm{e}^{2}\mathrm{g}(-x)$
3. $y=\mathrm{e}^{2}\mathrm{g}(-x)\rightarrow y=\mathrm{e}^{2}\mathrm{g}(-x)-\mathrm{e}^{2}$

Alternative 3:

$\mathrm{f}(x) = \mathrm{e}^{-x+2} - \mathrm{e}^2 = \mathrm{e}^2\left(\mathrm{e}^{-x} - 1\right)$

Let $\mathrm{g}(x) = \mathrm{e}^x$. Then, $\mathrm{f}(x) = \mathrm{e}^2\left(\mathrm{e}^{-x} - 1\right) = \mathrm{e}^2\left[\mathrm{g}\left(-x\right) - 1\right]$

1. Reflection in the *y*-axis.
2. Translation of 1 unit in the negative *y*-axis direction.
3. Scaling parallel to the *y*-axis by the scale factor e^2.

1. $y = \mathrm{g}(x) \to y = \mathrm{g}(-x)$
2. $y = \mathrm{g}(-x) \to y + 1 = \mathrm{g}(-x)$
   $y = \mathrm{g}(-x) - 1$
3. $y = \mathrm{g}(-x) - 1 \to \dfrac{y}{\mathrm{e}^2} = \mathrm{g}(-x) - 1$
   $y = \mathrm{e}^2\left[\mathrm{g}(-x) - 1\right]$

(b)(i)

We note that $\sin 2\left(x + \dfrac{\pi}{4}\right) = \sin\left(2x + \dfrac{\pi}{2}\right) = \sin 2x \cos\dfrac{\pi}{2} + \cos 2x \sin\dfrac{\pi}{2} = \cos 2x$.

Hence translation of $\dfrac{\pi}{4}$ units in the negative *x*-axis direction.

(ii)

We use the double angle formula to note that $\mathrm{g}(x) = \cos 2x = 1 - 2\sin^2 x$.

Let $\mathrm{h}(x) = \sin^2 x$. Then, $\mathrm{g}(x) = 1 - 2\sin^2 x = 1 - 2\mathrm{h}(x)$.

Hence

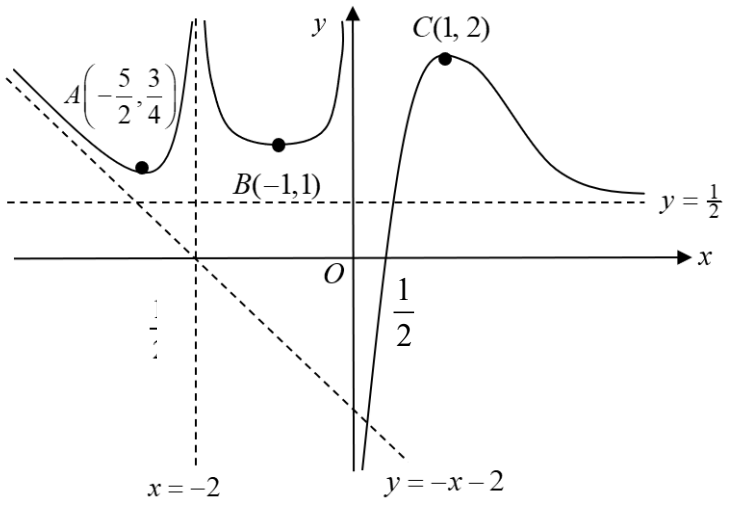
1. Reflection in the *x*-axis.
2. Scaling parallel to the *y*-axis by the scale factor 2.
3. Translation of 1 unit in the positive *y*-axis direction.

[Note that 1 and 2 are interchangeable]

1. $y = \mathrm{h}(x) \to -y = \mathrm{h}(x)$
   $y = -\mathrm{h}(x)$
2. $y = -\mathrm{h}(x) \to \dfrac{y}{2} = -\mathrm{h}(x)$
   $y = -2\mathrm{h}(x)$
3. $y = -2\mathrm{h}(x) \to y - 1 = -2\mathrm{h}(x)$
   $y = 1 - 2\mathrm{h}(x)$

4 DHS Prelim 9758/2022/01/Q2

The diagram below shows the graph of $y = \mathrm{f}(x)$. The curve crosses the x-axis at $x = \frac{1}{2}$ and has turning points at $A\left(-\frac{5}{2}, \frac{3}{4}\right)$, $B(-1,1)$ and $C(1,2)$. It has asymptotes $y = -x - 2$, $y = \frac{1}{2}$, $x = 0$ and $x = -2$.



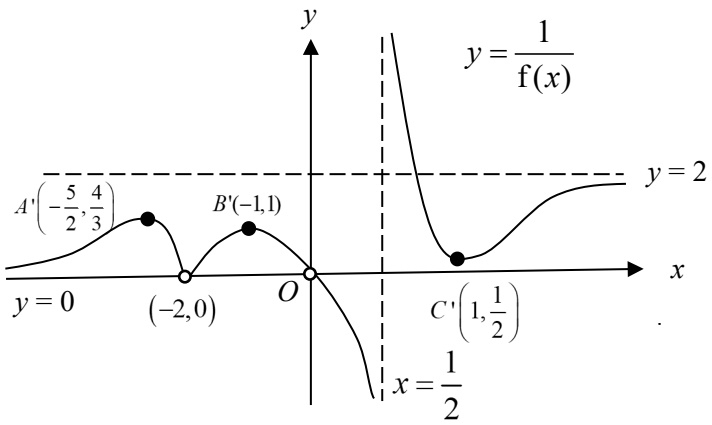
On separate diagrams, sketch the following graphs. In each case, show clearly the equations of any asymptotes, coordinates of any points of intersection with both axes and the points corresponding to A, B and C.

(a) $y = \dfrac{1}{\mathrm{f}(x)}$ [3]

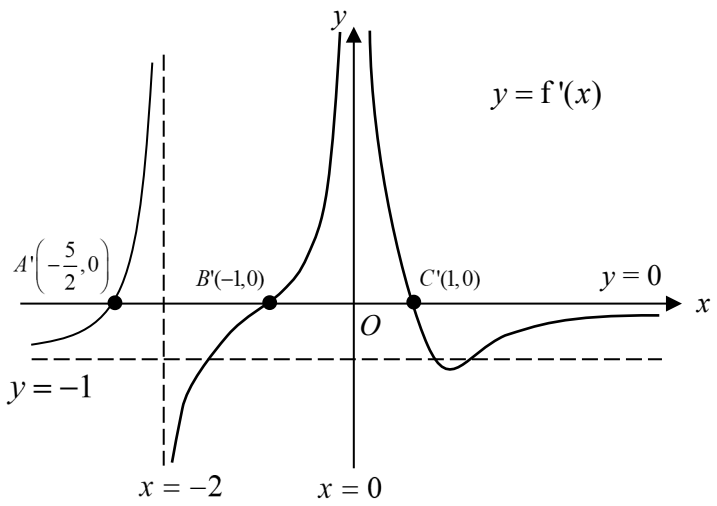
(b) $y = \mathrm{f}'(x)$ [3]

Solution:

(a)



(b)



5 ACJC Prelim 9758/2021/01/Q4(modified)

The curve C has equation $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = \dfrac{x(x+a)}{x-a}$ and a is a positive real constant.

(i) Find algebraically, in terms of a, the set of values that y can take. [4]

(ii) Sketch C, indicating clearly the equations of any asymptotes and the coordinates of the turning points and points where the curve crosses the axes. [4]

(iii) By adding a suitable graph to your same diagram in **(ii)** and labelling it clearly, solve the inequality $\dfrac{3x(x+a)}{x-a} < |x|$. [3]

Solution:

(i)

Consider $y = k$, where k is a constant

$$\frac{x(x+a)}{x-a} = k$$
$$x^2 + ax = xk - ak$$
$$x^2 + (a-k)x + ak = 0 \qquad (*)$$

For the range of values that y can take, the line $y = k$ and the curve C would have point(s) of intersection.

$$\therefore \text{Discriminant} \geq 0$$
$$(a-k)^2 - 4ak \geq 0$$
$$a^2 - 2ak + k^2 - 4ak \geq 0$$
$$k^2 - 6ak + a^2 \geq 0 \quad -----(1)$$

Consider $k^2 - 6ak + a^2 = 0$

$$k = \frac{6a \pm \sqrt{36a^2 - 4a^2}}{2} = \left(3 \pm 2\sqrt{2}\right)a$$

So, the solution to inequality (1) is $k \geq \left(3 + 2\sqrt{2}\right)a$ or $k \leq \left(3 - 2\sqrt{2}\right)a$

Hence, set of values of y is $\left(-\infty, \left(3 - 2\sqrt{2}\right)a\right] \cup \left[\left(3 + 2\sqrt{2}\right)a, \infty\right)$.

(ii)

We have $\mathrm{f}(x) = \dfrac{x(x+a)}{x-a} = x + 2a + \dfrac{2a^2}{x-a}$ and thus the oblique asymptote has equation $y = x + 2a$.

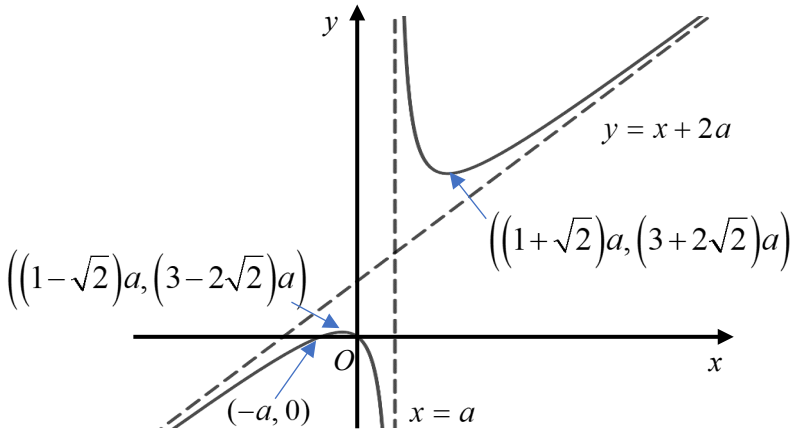
When $y = 0, x = -a$ or 0.

When $y=\left(3\pm2\sqrt{2}\right)a$, $x=\left(1\pm\sqrt{2}\right)a$ (solve via quadratic equation - DIY).

Otherwise, $\mathrm{f}'(x)=1-\dfrac{2a^2}{(x-a)^2}=0\Leftrightarrow x=a\pm\sqrt{2}a=(1\pm\sqrt{2})a$.

Or we can also note from (*) that when y takes on these values (*) has only repeated real roots, and sum of roots = –coefficient of x = $k-a$ which implies that

$$x=\frac{k-a}{2}=\frac{\left(3\pm2\sqrt{2}\right)a-a}{2}=\left(1\pm\sqrt{2}\right)a$$

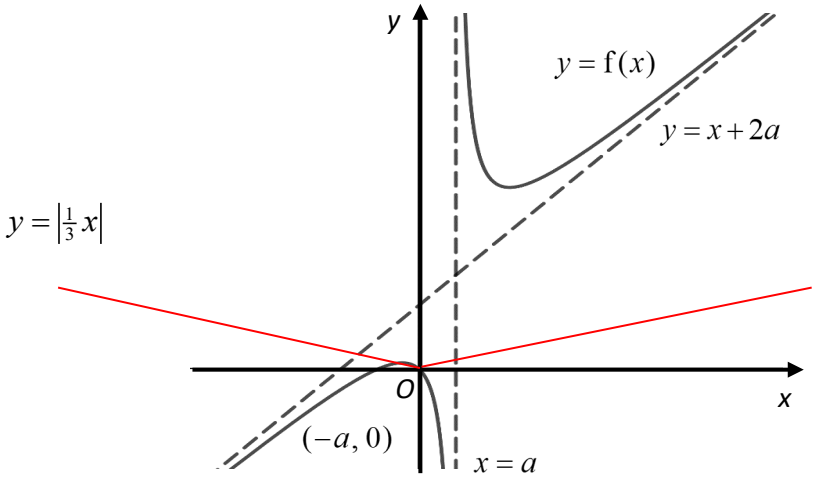


(iii)

$$\frac{3x(x+a)}{x-a}<|x|$$

$$\frac{x(x+a)}{x-a}<\frac{1}{3}|x|$$

We add in the graph of $y=\dfrac{1}{3}|x|=\left|\dfrac{1}{3}x\right|$ to the diagram.



Consider $\dfrac{x(x+a)}{x-a} = -\dfrac{1}{3}x$

$3x^2 + 3ax = -x^2 + ax$

$4x^2 + 2ax = 0$

$x(2x+a) = 0$

$x = 0$ or $x = -\frac{a}{2}$

Hence, $y = \left|\frac{1}{3}x\right|$ intersect the graph at $x = 0$ and $x = -\dfrac{a}{2}$. From the graph,

$$x < -\frac{a}{2} \text{ or } 0 < x < a\,.$$

Note :

$y = \frac{1}{3}x$ does not intersect the graph at a positive value of x since the gradient is less steep than that of the oblique asymptote.

6 HCI Prelim 9758/2021/01/Q9

Given that $y = (\tan x + \sec x)^2$, show that $\cos x \dfrac{\mathrm{d}y}{\mathrm{d}x} = 2y$. [2]

(i) By repeat differentiation, find the Maclaurin series for y up to and including the term in x^3. [4]

(ii) Using the result obtained in **(i)** estimate the value of $\tan 1^\circ + \sec 1^\circ$ to 4 decimal places. [2]

(iii) By expressing y in terms of sine and cosine, use the standard series in MF27 to find the series expansion for y up to and including the term in x^3. [3]

(iv) Comment on the results obtained from part **(i)** and part **(iii)**. [1]

Solution:

$y = (\tan x + \sec x)^2$

$\dfrac{\mathrm{d}y}{\mathrm{d}x} = 2(\tan x + \sec x)(\sec^2 x + \sec x \tan x)$

$\cos x \dfrac{\mathrm{d}y}{\mathrm{d}x} = 2(\tan x + \sec x)(\sec x + \tan x)$ ← Multiply both sides by cos x

$\cos x \dfrac{\mathrm{d}y}{\mathrm{d}x} = 2(\tan x + \sec x)^2$

$\cos x \dfrac{\mathrm{d}y}{\mathrm{d}x} = 2y$ (shown)

(i) Differentiate w.r.t x,

$$\cos x \frac{\mathrm{d}^2 y}{\mathrm{d}x^2} - \sin x \frac{\mathrm{d}y}{\mathrm{d}x} = 2\frac{\mathrm{d}y}{\mathrm{d}x}$$

Differentiate w.r.t x,

$$\cos x \frac{\mathrm{d}^3 y}{\mathrm{d}x^3} - \sin x \frac{\mathrm{d}^2 y}{\mathrm{d}x^2} - \sin x \frac{\mathrm{d}^2 y}{\mathrm{d}x^2} - \cos x \frac{\mathrm{d}y}{\mathrm{d}x} = 2\frac{\mathrm{d}^2 y}{\mathrm{d}x^2}$$

When $x = 0$, $y = 1, \dfrac{\mathrm{d}y}{\mathrm{d}x} = 2, \dfrac{\mathrm{d}^2 y}{\mathrm{d}x^2} = 4, \dfrac{\mathrm{d}^3 y}{\mathrm{d}x^3} = 10$

$$\therefore y = \mathrm{f}(0) + \mathrm{f}'(0)x + \frac{\mathrm{f}''(0)}{2}x^2 + \frac{\mathrm{f}'''(0)}{3!}x^3 + \ldots$$

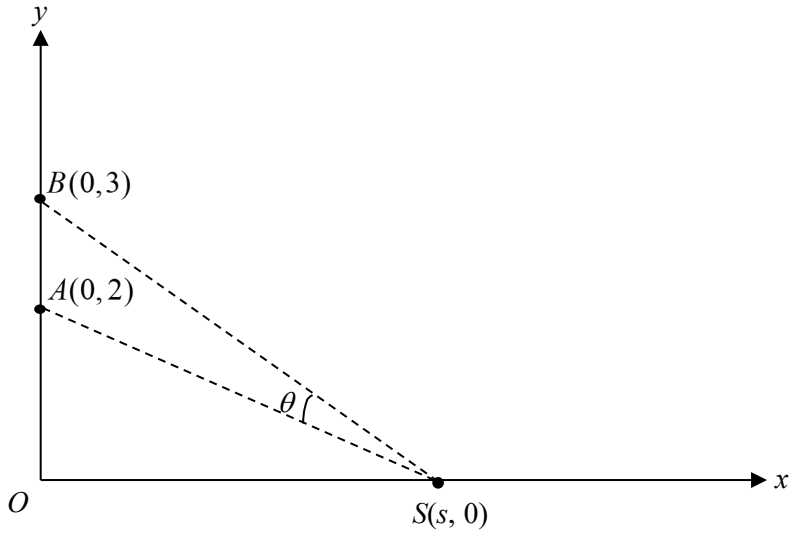
$$= 1 + 2x + 2x^2 + \frac{5}{3}x^3 + \ldots$$

(ii) Note that x in (i) is in radian.

$1^\circ = \dfrac{\pi}{180}$

	$\tan 1^\circ + \sec 1^\circ \approx \sqrt{1+2(\frac{\pi}{180})+2(\frac{\pi}{180})^2+\frac{5}{3}(\frac{\pi}{180})^3}$ $=\sqrt{1.03552468}$ $=1.0176(4\text{d.p.})$
(iii)	$y=\left(\tan x+\sec x\right)^2$ $=\left(\frac{\sin x+1}{\cos x}\right)^2$ $=(1+\sin x)^2(\cos x)^{-2}$ $\approx\left(1+x-\frac{x^3}{6}\right)^2\left(1-\frac{x^2}{2}\right)^{-2}$ $=\left[1+2\left(x-\frac{x^3}{6}\right)+\left(x-\frac{x^3}{6}\right)^2\right](1+x^2+...)$ $\approx(1+2x+x^2-\frac{1}{3}x^3)(1+x^2+...)$ $=1+2x+2x^2+\frac{5}{3}x^3+...$ From MF27, $\sin x=x-\frac{x^3}{3!}+\frac{x^5}{5!}-\ldots+\frac{(-1)^r x^{2r+1}}{(2r+1)!}+\ldots$ (all x) $\cos x=1-\frac{x^2}{2!}+\frac{x^4}{4!}-\ldots+\frac{(-1)^r x^{2r}}{(2r)!}+\ldots$ (all x)
(iv)	We expect the answers from both parts to be the same and the results have proven so.

7 In the diagram below, a soccer player, represented by the point $S(s,0)$, is running with a ball along the touchline, represented by the x-axis. The goalposts are represented by the points $A(0,2)$ and $B(0,3)$ and $\theta=\angle ASB$. It is assumed that $s>0$.



(i) By applying the cosine rule, show that $\cos^2\theta=\dfrac{(s^2+6)^2}{(s^2+4)(s^2+9)}$. [2]

(ii) Let $\mathrm{f}(s)=\dfrac{(s^2+6)^2}{(s^2+4)(s^2+9)}$. Express $\mathrm{f}(s)$ in the form $P+\dfrac{Q}{s^2+4}+\dfrac{R}{s^2+9}$, where P, Q and R are constants to be determined. Hence find $\mathrm{f}'(s)$. [4]

(iii) The soccer player decides to shoot when θ is maximum. Using your result from **(ii),** determine the exact value of s at which the soccer player should shoot, justifying your answer. [4]

Solution:

7 (i)

Applying the cosine rule to ΔASB, we have

$$(3-2)^2=\left(\sqrt{2^2+s^2}\right)^2+\left(\sqrt{3^2+s^2}\right)^2-2\sqrt{2^2+s^2}\sqrt{3^2+s^2}\cos\theta.$$

Rearranging we have

$$\cos\theta=\frac{\left(4+s^2\right)+\left(9+s^2\right)-1}{2\sqrt{4+s^2}\sqrt{9+s^2}}=\frac{s^2+6}{\sqrt{4+s^2}\sqrt{9+s^2}},$$

which upon squaring both sides gives

$$\cos^2\theta=\frac{(s^2+6)^2}{(s^2+4)(s^2+9)}.$$

(ii)

$$\mathrm{f}(s)=\frac{(s^2+6)^2}{(s^2+4)(s^2+9)}.$$

Since the coefficient of s^4 in both the numerator and denominator of f(s) is 1, then the value of P is 1.

We have $\dfrac{(s^2+6)^2}{(s^2+4)(s^2+9)}=1+\dfrac{Q}{s^2+4}+\dfrac{R}{s^2+9}$, Multiply both sides by $(s^2+4)(s^2+9)$

so

$$(s^2+6)^2=(s^2+4)(s^2+9)+Q(s^2+9)+R(s^2+4).$$

Comparing coefficients gives

Constants: $\quad 36=36+9Q+4R \quad\Rightarrow\quad 9Q+4R=0$

s^2: $\quad 12=13+Q+R \quad\Rightarrow\quad Q+R=-1,$

Solving, $Q=\dfrac{4}{5}$ and $R=-\dfrac{9}{5}$.

Thus, $\quad \mathrm{f}(s)=1+\dfrac{1}{5}\left(\dfrac{4}{s^2+4}-\dfrac{9}{s^2+9}\right)$

And $\quad \mathrm{f}'(s)=\dfrac{2s}{5}\left(\dfrac{9}{(s^2+9)^2}-\dfrac{4}{(s^2+4)^2}\right).$

(iii)

When $\mathrm{f}'(s)=0$, we have

$$\frac{2s}{5}\left(\frac{9}{(s^2+9)^2}-\frac{4}{(s^2+4)^2}\right)=0$$

$$\frac{9}{(s^2+9)^2}-\frac{4}{(s^2+4)^2}=0 \quad (\text{as } s>0)$$

$$\frac{9}{(s^2+9)^2}=\frac{4}{(s^2+4)^2}$$

$$\frac{3}{s^2+9}=\frac{2}{s^2+4} \quad (\text{as all quantities are positive})$$

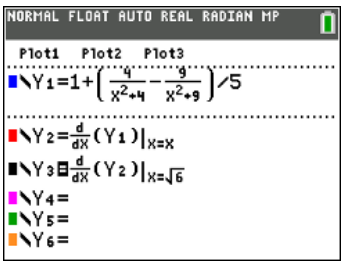
Thus $s^2=6$, giving $s=\sqrt{6}$ as $s>0$.

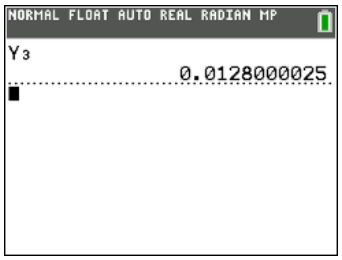
To check nature of stationary value,

Method 1: Second derivative test

Using GC, $\mathrm{f}''\left(\sqrt{6}\right)=0.0128>0$ therefore $\mathrm{f}(s)$ is minimum when $s=\sqrt{6}$.

That is, $\cos^2\theta$ is minimum when $s=\sqrt{6}$.





Method 2: First derivative test

$$\mathrm{f}'(s)=\frac{2s}{5}\left(\frac{9}{(s^2+9)^2}-\frac{4}{(s^2+4)^2}\right)=\frac{2s}{5}\left(\frac{3^2(s^2+4)^2-2^2(s^2+9)^2}{(s^2+9)^2(s^2+4)^2}\right)$$

$$=\frac{2s}{5}\left(\frac{(5s^2+30)(s^2-6)}{(s^2+9)^2(s^2+4)^2}\right)=\frac{2s(s^2+6)(s+\sqrt{6})(s-\sqrt{6})}{(s^2+9)^2(s^2+4)^2}$$

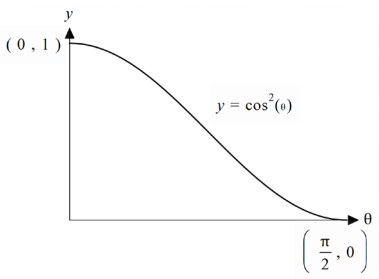
This implies that

s	$\sqrt{6}^-$	$\sqrt{6}$	$\sqrt{6}^+$
$(s-\sqrt{6})$	<0	0	>0
$\mathrm{f}'(s)$	<0	0	>0
$\mathrm{f}(s)$	$\searrow$	$__$	$\nearrow$

Thus, $\mathrm{f}(s)$ is minimum when $s=\sqrt{6}$.

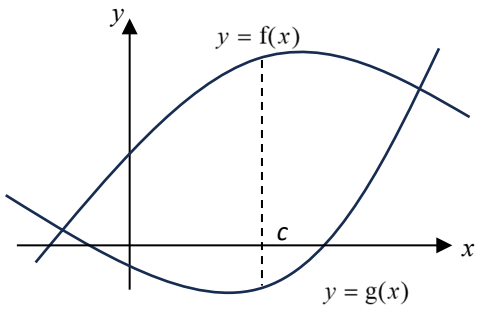
That is, $\cos^2\theta$ is minimum when $s=\sqrt{6}$.

Since $\cos^2\theta$ is decreasing for $\theta\in\left[0,\frac{\pi}{2}\right]$, when f is minimum, $\cos^2\theta$ is minimum and this implies θ is maximum. Hence the soccer player should shoot when $s=\sqrt{6}$.



8 ACJC Prelim 9758/2023/01/Q9

(a) The following diagram shows the graphs of two differentiable functions $y = \mathrm{f}(x)$ and $y = \mathrm{g}(x)$. At $x = c$, the distance between the two functions is a maximum. Show that $\mathrm{f}'(c) = \mathrm{g}'(c)$ where $\mathrm{f}'(x)$ and $\mathrm{g}'(x)$ are derivatives of $\mathrm{f}(x)$ and $\mathrm{g}(x)$ respectively. Using the second derivative, explain clearly why the distance is a maximum. [3]



(b) Two runners Andy and Ben, with different strategies and abilities, participate in a 10000 m race. The distance in metres ran in x minutes by Andy and Ben are given by the functions $\mathrm{A}(x)$ and $\mathrm{B}(x)$ respectively as defined below.

$$\mathrm{A}(x) = \begin{cases} -0.16x^3 + 12x^2, & 0 \le x \le 50 \\ 10000, & x > 50 \end{cases}.$$

$$\mathrm{B}(x) = 5000\log_3(x+9) - 10000, \ 0 \le x \le 72.$$

(i) Sketch the graphs of $y = \mathrm{A}(x)$ and $y = \mathrm{B}(x)$ on the same diagram for $0 \le x \le 72$. [2]

(ii) Andy and Ben are furthest apart at $x = c$. By using **(a)** or otherwise, show that $x = c$ satisfies the equation $Px^2 + Qx = \dfrac{R}{x+9}$, where P, Q and R are constants to be determined, and find the value(s) of c. [4]

(iii) Hence or otherwise, find the furthest distance between Andy and Ben and state who was leading at that moment. [2]

Solution:

8(a)

Let $D = \mathrm{f}(x) - \mathrm{g}(x)$.

$$\frac{\mathrm{d}D}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{f}(x) - \mathrm{g}(x)\right) = \mathrm{f}'(x) - \mathrm{g}'(x)$$

At $x = c$, distance is maximum, i.e.

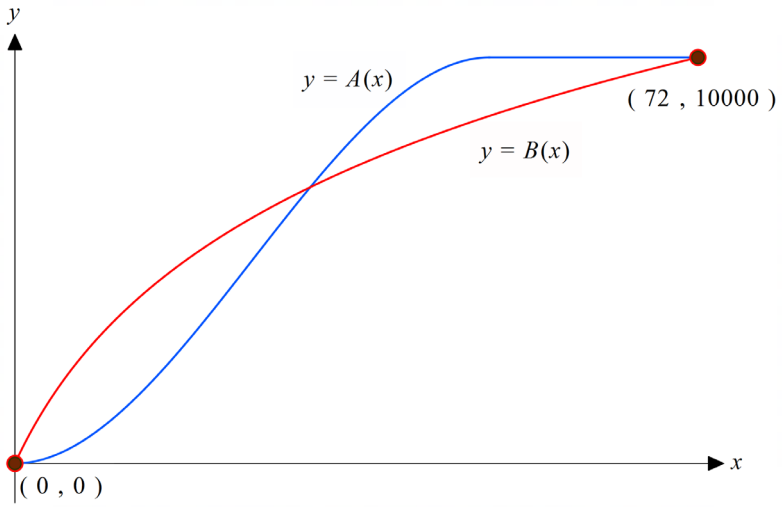
$$\left.\frac{\mathrm{d}D}{\mathrm{d}x}\right|_{x=c} = 0 \Rightarrow \mathrm{f}'(c) - \mathrm{g}'(c) = 0 \Rightarrow \mathrm{f}'(c) = \mathrm{g}'(c) \text{ (shown)}$$

From the diagram, at $x = c$, $y = \mathrm{f}(x)$ is concave downwards, i.e. $\mathrm{f}''(c) < 0$ while $y = \mathrm{g}(x)$ is concave upwards, i.e. $\mathrm{g}''(c) > 0$

Therefore at $x = c$, $\dfrac{\mathrm{d}^2 D}{\mathrm{d}x^2} = \mathrm{f}''(c) - \mathrm{g}''(c) < 0$.

Hence the distance is a maximum.

(b) (i)



(b) (ii)

From **(a)**, we know that the maximum will occur when $\mathrm{A}'(c) = \mathrm{B}'(c)$. Thus, c will satisfy the equation $\mathrm{A}'(x) = \mathrm{B}'(x)$.

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(-0.16x^3 + 12x^2\right) = \frac{\mathrm{d}}{\mathrm{d}x}\left(5000\log_3(x+9) - 10\,000\right)$$

$$-0.48x^2 + 24x = \frac{5000}{(x+9)\ln 3}$$

Therefore $P = -0.48$, $Q = 24$, $R = \dfrac{5000}{\ln 3}$.

Using GC to solve the equation gives

$x = 11.906\,(5\text{ s.f.}) = 11.9\,(3\text{ s.f.})$

$x = 46.296\,(5\text{ s.f.}) = 46.3\,(3\text{ s.f.})$

Note that:

$$\frac{\mathrm{d}}{\mathrm{d}x}(\log_a x) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\ln x}{\ln a}\right), \text{using change of base}$$

$$= \frac{1}{\ln a}\frac{\mathrm{d}}{\mathrm{d}x}(\ln x)$$

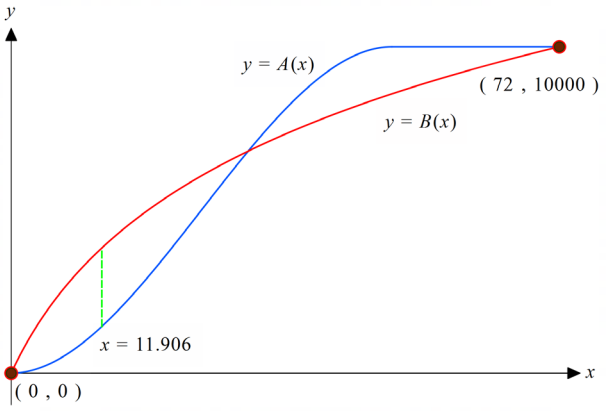
$$= \frac{1}{x\ln a}$$

(b) (iii)

$\mathrm{A}(11.906) - \mathrm{B}(11.906) = -2404.8$

$\mathrm{A}(46.296) - \mathrm{B}(46.296) = 1580.9$

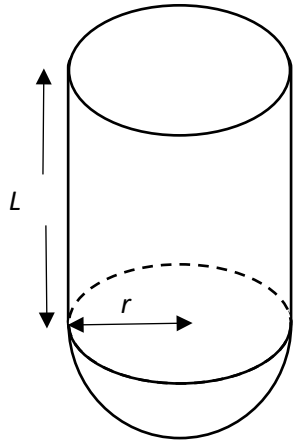
Therefore the furthest distance between them is 2404.8m (or 2400m) and Ben is in front of Andy.



9 DHS Prelim 9758/2025/P1/Q11

[The surface area and volume of a sphere are given by $4\pi r^2$ and $\frac{4}{3}\pi r^3$ respectively.]

An aquarist who has a fixed budget of \$$k$ wants to make a goldfish sanctuary. His design of the tank consists of a circular cover, a cylinder of length L cm and a hemispherical base of radius r cm neatly fitted together. The costs of the material per cm^2 used for the cover, the cylindrical surface and hemispherical base are \$20, \$10 and \$30 respectively. Assume that the tank is made of material with negligible thickness.

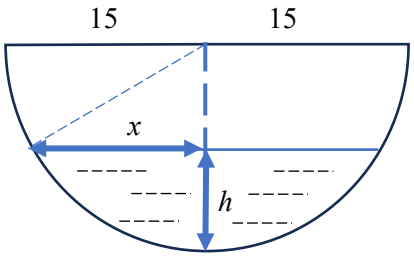


(a) Show that the volume of the tank is V cm^3, where $V = \frac{k}{20}r - \frac{10}{3}\pi r^3$. [3]

(b) As r varies, find the cost of the material used to make the hemispherical base in terms of k when V is at its maximum value. You need to show that V is a maximum. [3]

The aquarist has fixed the radius of the cylinder to be 15 cm. Initially, there is some water in the hemispherical part of the tank. However, due to a defect in the tank, water is leaking at a constant rate of 20 cm^3 per minute.

The depth of the water at the hemispherical bottom is denoted by h cm, and the radius of the water surface is x cm. The volume of water in the hemispherical bottom is given by W cm^3, where $W = \frac{\pi h\left(3x^2 + h^2\right)}{6}$.



(c) By formulating a relationship between x and h, find, when $h = 3$, the rate of change of

(i) the depth of water, [3]

(ii) the radius of the water surface. [3]

Solutions

9(a)

$$\begin{aligned} k &= 2\pi r^2(30) + 2\pi rL(10) + \pi r^2(20) \\ &= 80\pi r^2 + 20\pi rL \\ &= 20\pi r(4r + L) \end{aligned}$$

$$\Rightarrow L = \frac{k}{20\pi r} - 4r$$

$$\begin{aligned} V &= \frac{2}{3}\pi r^3 + \pi r^2 L \\ &= \frac{2}{3}\pi r^3 + \pi r^2\left(\frac{k}{20\pi r} - 4r\right) \\ &= \frac{2}{3}\pi r^3 + \frac{kr}{20} - 4\pi r^3 \\ &= \frac{k}{20}r - \frac{10}{3}\pi r^3 \text{ (shown)} \end{aligned}$$

(b)

$$\frac{\mathrm{d}V}{\mathrm{d}r} = \frac{k}{20} - 10\pi r^2$$

For maximum V as r varies, $\dfrac{\mathrm{d}V}{\mathrm{d}r} = 0$

$$\frac{k}{20} - 10\pi r^2 = 0$$

$$r = \sqrt{\frac{k}{200\pi}} \quad (\text{reject } r < 0)$$

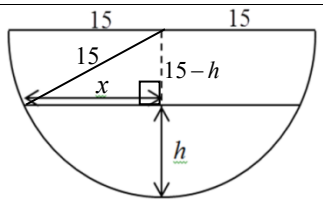
$$\frac{\mathrm{d}^2V}{\mathrm{d}r^2} = -20\pi r < 0$$

Therefore, V is maximum when $r = \sqrt{\dfrac{k}{200\pi}}$.

Cost of material for the hemispherical base

$$= 2\pi r^2(30) = 2\pi\left(\sqrt{\frac{k}{200\pi}}\right)^2(30) = \frac{3k}{10} \text{ dollars}$$

(c)(i)



By Pythagoras' Theorem, $15^2 = x^2 + (15-h)^2$

$$\begin{aligned} x^2 &= 15^2 - (15-h)^2 \\ &= (15-(15-h))(15+(15-h)) \\ &= h(30-h) = 30h - h^2 \end{aligned}$$

	$W = \dfrac{\pi h\left(3x^2 + h^2\right)}{6}$ $= \dfrac{\pi h\left[3\left(30h - h^2\right) + h^2\right]}{6}$ $= \dfrac{\pi h\left(90h - 2h^2\right)}{6}$ $= \dfrac{\pi\left(45h^2 - h^3\right)}{3}$ $\dfrac{\mathrm{d}W}{\mathrm{d}h} = \dfrac{\pi\left(90h - 3h^2\right)}{3} = \pi\left(30h - h^2\right)$ Given: $\dfrac{\mathrm{d}W}{\mathrm{d}t} = -20 \text{ cm}^3 / \text{min}$ So, $\dfrac{\mathrm{d}h}{\mathrm{d}t} = \dfrac{\mathrm{d}h}{\mathrm{d}W} \times \dfrac{\mathrm{d}W}{\mathrm{d}t} = \dfrac{1}{\pi\left(30h - h^2\right)} \times (-20)$ At $h = 3$, $\dfrac{\mathrm{d}h}{\mathrm{d}t} = \dfrac{-20}{\pi\left(30(3) - (3)^2\right)} = -0.078595 = -0.0786 \text{ cm/min } (3 \text{ s.f.})$
(c)(ii)	$x^2 = 30h - h^2$ Differentiate the above equation implicitly with respect to t, $2x\dfrac{\mathrm{d}x}{\mathrm{d}t} = 30\dfrac{\mathrm{d}h}{\mathrm{d}t} - 2h\dfrac{\mathrm{d}h}{\mathrm{d}t}$ ----- (1) When $h = 3$, $x = \sqrt{30h - h^2} = \sqrt{90 - 9} = 9$ Subst $x = 9$, $h = 3$ and $\dfrac{\mathrm{d}h}{\mathrm{d}t} = -0.078595$ into (1), $2(9)\dfrac{\mathrm{d}x}{\mathrm{d}t} = 30\left(-0.078595\right) - 2(3)\left(-0.078595\right)$ $\dfrac{\mathrm{d}x}{\mathrm{d}t} = -0.10479 = -0.105 \text{ cm / min } (3\text{s.f.})$